

1. Jalankan MASTERDATA.sql

Ini hanya dilakukan **sekali saja**.

Script ini membuat seluruh environment:

```
WAREHOUSE_SAMPLE
|
├─ dbo
|   ├── Customers
|   ├── Products
|   ├── Employees
|   ├── Orders
|   └─ OrderDetails
|
├─ STG
|   ├── Customer_Staging
|   ├── Product_Staging
|   ├── Employee_Staging
|   └─ Sales_Staging
|
├─ DW
|   ├── DimCustomer
|   ├── DimProduct
|   ├── DimEmployee
|   ├── FactSales
|   └─ AggSalesMonthly
|
└─ ETL tables
    ├── ETL_Metadata
    └─ ETL_Log
```

Setelah ini database **sudah siap menerima ETL**.

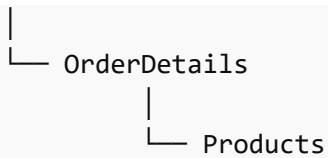
2. Isi data di OLTP source

Minimal isi tabel ini:

```
dbo.Customers
dbo.Products
dbo.Employees
dbo.Orders
dbo.OrderDetails
```

Contoh relasinya:

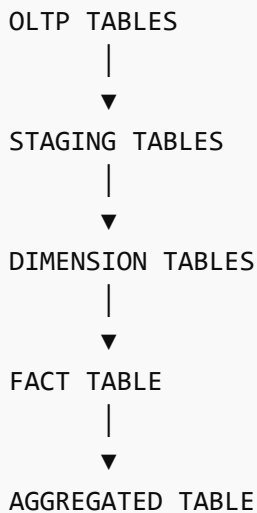
```
Customers
|
└─ Orders
```



Employee juga terhubung ke `Orders` .

3. Jalankan `ETL_Pipeline.sql`

Script ini akan melakukan proses berikut secara otomatis:



Secara teknis prosesnya:

1. Ambil data baru (`LastModified`)
2. Load ke **STG**
3. Merge ke **DimCustomer**
4. Merge ke **DimProduct**
5. Merge ke **DimEmployee**
6. Load **FactSales**
7. Hitung **AggSalesMonthly**
8. Update metadata ETL

4. Tableau Connect ke Warehouse

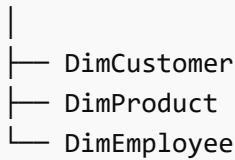
Di Tableau Anda tidak perlu connect ke tabel OLTP.

Connect langsung ke:

```
DW.FactSales
DW.DimCustomer
DW.DimProduct
DW.DimEmployee
```

Relasinya di Tableau:

FactSales



Ini disebut **Star Schema**.

5. Contoh Dashboard yang Bisa Anda Buat

Begitu connect ke Tableau, Anda bisa langsung membuat:

Sales by Product

ProductName
SUM(SalesAmount)

Sales by Country

Country
SUM(SalesAmount)

Sales by Employee

EmployeeName
SUM(SalesAmount)

Monthly Sales Trend

OrderDate (Month)
SUM(SalesAmount)

Top 10 Products

ProductName
SUM(SalesAmount)
Top N Filter

6. Cara Test Incremental ETL

Masukkan order baru di OLTP:

```
INSERT INTO Orders  
VALUES (1001, 1, 1, GETDATE(), GETDATE());
```

Lalu jalankan lagi:

```
ETL_Pipeline.sql
```

Hasilnya:

FactSales akan bertambah

Ini meniru **real near real-time ETL pipeline**.

Kesimpulan

Ya. Untuk belajar Tableau dengan SQL Server, cukup:

1 Jalankan

```
MASTERDATA.sql
```

2 Isi data di

Customers
Products
Employees
Orders
OrderDetails

3 Jalankan

```
ETL_Pipeline.sql
```

4 Connect Tableau ke:

DW.FactSales

dan join ke dimension.

Itu sudah cukup untuk membuat dashboard BI yang realistis.